

Water Resources Research®



RESEARCH ARTICLE

10.1029/2025WR041518

Special Collection:

Integrating In Situ, Remote Sensing, And Physically Based Modeling Approaches to Understand Global Freshwater Ice Dynamics

Key Points:

- Satellite-derived floe size parameterization reduces ice model biases in the Great Lakes
- Replacing constant floe size with spatially-varying climatology boosts ice thickness prediction accuracy by over 25%
- Operational models using variable floe sizes result in more rapid spring ice melt

Supporting Information:

Supporting Information may be found in the online version of this article.

Correspondence to:

K. Pereira,
khuele@mines.edu

Citation:

Pereira, K., Anderson, E. J., Kessler, J., & Fujisaki-Manome, A. (2026). Integrating ice analyses and varying floe size parameterization into Great Lakes Ice prediction. *Water Resources Research*, 62, e2025WR041518. <https://doi.org/10.1029/2025WR041518>

Received 30 JUN 2025

Accepted 1 JUN 2026

Integrating Ice Analyses and Varying Floe Size Parameterization Into Great Lakes Ice Prediction

Kaitlin Pereira¹ , Eric J. Anderson¹ , James Kessler² , and Ayumi Fujisaki-Manome³ 

¹Civil & Environmental Engineering, Colorado School of Mines, Golden, CO, USA, ²Great Lakes Environmental Research Laboratory, National Oceanic and Atmospheric Administration, Ann Arbor, MI, USA, ³Cooperative Institute for Great Lakes Research, University of Michigan, Ann Arbor, MI, USA

Abstract Accurate prediction of Great Lakes ice cover is critical for regional weather, navigation support, and safety. Current operational models often exhibit biases, such as over-prediction of ice concentration and delayed spring melt, potentially due to simplified parameterizations. This study addresses this issue by first characterizing ice floe size variability using 12 years (2010–2021) of satellite-derived ice charts from the U.S. National Ice Center (NIC). We tested spatially- and temporally-varying floe size parameterizations in the Finite Volume Community Ocean Model (FVCOM), comparing its performance against the standard operational configuration, which uses a constant 300 m floe size. The analysis reveals that floe sizes are highly variable and correlate with winter severity. Model simulations for a high-ice (2019) and low-ice (2020) winter show that variable floe size parameterizations significantly reduce model bias in ice concentration and improve categorical accuracy for ice thickness when compared to a constant floe size configuration. For example, exact categorical accuracy for Lake Superior ice thickness improved from 41% to 67% in 2019. These findings demonstrate that integrating variable ice floe size from satellite data resolves a key model deficiency and offers a practical path toward improving operational ice forecasting in the Great Lakes.

Plain Language Summary Current ice prediction models for the Great Lakes use a fixed ice floe size of 300 m, but this does not reflect the real variability in floe sizes observed across the lakes. This simplification may lead to errors, especially by predicting that ice will melt too slowly in the spring. In this study, we analyzed 12 years of satellite data (2010–2021) to see how ice floe sizes vary across the Great Lakes. We found that ice floe sizes change depending on the location and severity of the winter. In mild winters, smaller floes (less than 100 m) are more common, while in colder, more severe winters, larger floes (500–2,000 m) appear more often. When these variable ice floe sizes were implemented in a computer model, the accuracy of predicted ice thickness and concentration improved significantly. This work shows that including satellite-derived ice data in computer forecast models can help provide better forecasts for the Great Lakes.

1. Introduction

The Laurentian Great Lakes (hereafter Great Lakes) make up the largest surface freshwater system on Earth, consisting of approximately 84% of the freshwater supply in the United States and 21% of the world's freshwater supply (Shiklomanov, 1993). Located along the United States and Canadian border, the five Great Lakes and their connecting waterways contain more than 7,000 km of coastline (Woolway et al., 2021) and 244,160 km² of surface area (Figure 1).

In general, ice formation in the Great Lakes begins in early December in shallow bays and along coastlines, reaching a maximum concentration in late February, and melting away in April or May. However, recent studies have indicated a decline in ice concentration over the last 40 years, with a high degree of interannual variability that is consistent with teleconnection patterns (Lin, Fujisaki-Manome, & Wang, 2022; Wang et al., 2012, 2018). Ice conditions in the lakes are critical to a number of stakeholders through direct impacts to commercial navigation, government and private ice breaking activities, search and rescue operations, and contaminant or oil spill response, as well as through indirect means such as weather modulation in the region. Early versions of weather and climate models included poor representations of the Great Lakes, but recent work has demonstrated the importance of accurate representation of lake surface temperature and ice concentration to forecasts of lake-effect precipitation, turbulent heat fluxes, and other meteorological parameters (Fujisaki-Manome, Mann, et al., 2020; Wright et al., 2013; Yeo et al., 2024).

© 2026. The Author(s).

This is an open access article under the terms of the [Creative Commons Attribution License](https://creativecommons.org/licenses/by/4.0/), which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

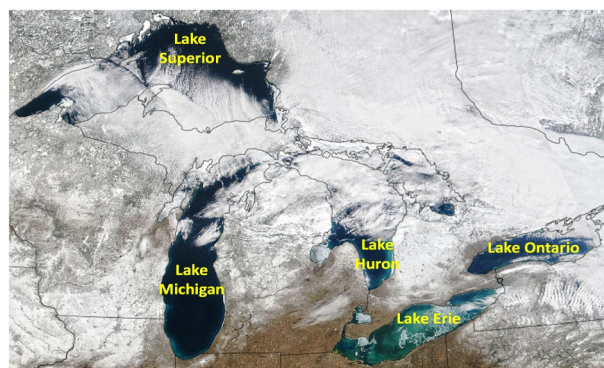


Figure 1. NASA Landsat satellite image showing ice cover across the Great Lakes. Adapted from the *NASA Visible Earth: Landsat Image Gallery*.

Due to the importance of ice information to stakeholders and weather forecasting, significant effort has been applied to ice prediction in the Great Lakes (Anderson et al., 2018; Bai et al., 2020; Cannon et al., 2023; Durnford et al., 2018; Fujisaki et al., 2012; Fujisaki-Manome, Anderson, et al., 2020; Lin, Fujisaki-Manome, & Anderson, 2022; Wake & Rumer Jr, 1983; Wang et al., 2010; Xue et al., 2017), including the development of operational nowcast and forecast models of ice conditions by the National Oceanic and Atmospheric Administration (NOAA; Anderson et al., 2018) and Environment and Climate Change Canada (ECCC; Durnford et al., 2018). In recent years, validation of modeled ice predictions has generally come from comparisons with ice charts produced by the United States National Ice Center (USNIC) and the Canadian Ice Service (CIS), together known as the North American Ice Service (NAIS), which provide maps of ice concentration, ice type, ice type inferred thickness, and floe size on a daily basis. While skill assessment of operational forecast models has demonstrated the capability of

mechanistic hydrodynamic-ice models to track observed seasonal ice conditions, evaluations are typically limited to predictions of ice concentration (Anderson et al., 2018).

A frequent performance deficiency in the simulations of Great Lakes ice concentration appears in the late spring, when ice is observed to melt rapidly while modeled concentration often lags behind the apparent rate of melting depicted in the ice charts. Some suggestions for this discrepancy include the current lack of ice-wave interaction in the operational models, insufficient mixing in the water column, and on-ice precipitation impacts, each of which has been demonstrated to play an important role in lake thermal structure and the occurrence of over-prediction of ice thickness (Bai et al., 2020; Fujisaki-Manome, Anderson, et al., 2020; Ye et al., 2019). However, the apparent model lag in the rate of ice melt in the spring may also stem from model parameterizations that govern ice thermodynamics and evolution. For instance, in the existing NOAA Great Lakes Operational Forecasting System (GLOFS; Anderson et al., 2018), the operational configuration employs a set of ice parameterizations that prescribe static values for ice strength, ridging, albedo, thickness distribution, light attenuation, and floe diameter. In particular, ice floe size or diameter is prescribed as a constant of 300 m. This default value was originally established for larger-scale sea ice predictions primarily covering pack ice regions; however, its validity for large lakes, and specifically Great Lakes, whose ice conditions resemble marginal ice zones, has not been investigated. Accurate representation of floe size in the Great Lakes may be critical for ice model performance as previous work has shown floe size to have a major influence on melt rates (Bateson et al., 2020).

Here, we use daily ice analyses from the coordinated US and Canadian Ice Centers to assess the spatio-temporal variability in Great Lakes ice floe size and evaluate the impact of employing satellite-derived floe sizes on ice prediction. We use a hydrodynamic-ice model similar to what is used in operations to test the constant ice floe size case, which is the existing operational configuration, against spatially- and temporally-variable ice floe sizes for two winter seasons. Model simulations are assessed for predictive accuracy of surface water temperature, ice concentration, and ice thickness using daily satellite-derived products. Overall, this effort aims to improve model parameterizations and ice prediction in Earth's largest lakes, where dynamic ice conditions dominate the winter season, and demonstrate the potential for integrating lake ice remote sensing into forecast operations.

2. Methods

2.1. Satellite-Derived Ice and Temperature Data

Satellite-derived ice products from the North American Ice Service (NAIS) are used to characterize ice floe size and assess model performance for ice concentration and ice thickness. Daily analyses contain georeferenced values for concentration, stage (proxy for thickness), and form (proxy for floe size) across the Great Lakes (USNIC). The synthesized products are developed using a combination of synthetic aperture radar (SAR; e.g., Radarsat, Sentinel-1), infrared and optical imagery (e.g., VIIRS, MODIS, GOES), and other products including ancillary data sources for relevant meteorological conditions when necessary. Ice data is provided in shapefile and gridded (1.8 km resolution) formats by the U.S. National Ice Center (<https://usicecenter.gov/>). Ice concentrations are given as values from 0% to 100%, and ice floe sizes, which are available from 2010-present, are binned into categories of <2 m, 2–20 m, 20–100 m, 100–500 m, 500–2,000 m, 2–10 km, and >10 km (Table 1). Ice thickness

Table 1

Classification of Ice Floe Sizes From the US National Ice Center (NIC) Data

Category	Description	Size range
1	small_cakes/brash	<2 m
2	cake	2–20 m
3	small_floe	20–100 m
4	med_floe	100–500 m
5	big_floe	500–2,000 m
6	vast_floe	2–10 km
7	giant_floe	>10 km

is not directly measured but instead is inferred from an ice type classification and given in categories of <100 mm, 100–150 mm, 150–300 mm, 300–700 mm, and >700 mm (Table 2). As a result, ice thickness is considered for the work presented here as a point of reference for model comparison.

Lake surface temperature (LST) data are acquired from the NOAA Great Lakes Surface Environmental Analysis (GLSEA3; Schwab et al., 1999). The NOAA GLSEA3 uses the Advanced Clear-sky Processor for Oceans (ACSPO) L3S-LEO SST, which is the level 3 Super Collated data from Low Earth Orbit (LEO) satellites (e.g., NPP, N20, MetOp A/B/C). Daily LST from the GLSEA3 are given on the same 1.8 km grid as supplied by the NAIS for cloud-free areas of the lakes. For cloudy areas, the GLSEA3 uses spatio-temporal interpolation to fill any missing data. The GLSEA3 is widely used for Great Lakes LST analysis and model validation (Anderson et al., 2018, 2021, 2024; Woolway et al., 2021; Yeo et al., 2024).

2.2. Hydrodynamic-Ice Model

The Finite Volume Community Ocean Model (FVCOM v4.3; Chen et al., 2006) is an unstructured-grid three-dimensional oceanographic model that has been adapted for freshwater applications and implemented for several Great Lakes studies (Anderson et al., 2010, 2015; Anderson & Schwab, 2017; Bai et al., 2013; Yeo et al., 2024). In particular, it is the basis for the NOAA Great Lakes Operational Forecast System (GLOFS; Anderson et al., 2018), a real-time forecast system that provides three-dimensional hydrodynamic and ice predictions across four different models that cover Lake Superior, Lake Michigan–Huron, Lake Erie, and Lake Ontario, respectively. Here we use an implementation of FVCOM that is similar to the GLOFS and has been employed in several research studies of the Great Lakes (Anderson et al., 2018; Yeo et al., 2024). Model horizontal resolution ranges from 30 m in the nearshore or near complex topography to 2.5 km in offshore regions, though most of the coastline has a resolution of approximately 200 m. Model bathymetry is interpolated from 3 arc-second bathymetric data from the NOAA National Centers for Environmental Information (NCEI). In the vertical direction, the models use a sigma- (terrain-following) vertical coordinate system with 21 uniformly spaced layers. Vertical diffusion is calculated using the Mellor–Yamada Level-2.5 turbulence closure scheme (MY-2.5; Mellor & Blumberg, 2004; Mellor & Yamada, 1982), and horizontal diffusion is prescribed by the Smagorinsky parameterization (Smagorinsky, 1963). Turbulent heat fluxes are calculated using the Coupled Ocean–Atmosphere Response Experiment algorithm (COARE 3.0; Fairall et al., 1996; Fairall et al., 2003; Edson et al., 2013).

FVCOM is internally-coupled with an unstructured grid version of the Los Alamos Sea Ice Model (CICE; Hunke & Dukowicz, 1997; Hunke et al., 2017; Gao et al., 2011). The unstructured grid CICE model inside of FVCOM

has been adapted for freshwater ice simulation and has been successfully applied to the Great Lakes real-time prediction and other studies (Anderson et al., 2018; Fujisaki-Manome, Anderson, et al., 2020; Ye et al., 2019; Yeo et al., 2024). CICE simulates ice dynamics and thermodynamics using an elastic-viscous-plastic rheology for internal stresses, and includes a five-category (5, 25, 65, 125, 205 cm) ice thickness distribution model that represents the sub-grid scale distribution of ice in response to thermodynamic and mechanical forcing (Thorndike et al., 1975). The change in ice volume in CICE is represented as

$$\frac{d(AH)}{dt} = A \left[w_{\text{top}} + w_{\text{bot}} + \frac{\pi H}{\alpha L} w_{\text{lat}} \right] \quad (1)$$

where A is the ice surface area, H is the ice thickness, L is the ice floe diameter, α is a constant ($=0.66$), and where w_{top} , w_{bot} , and w_{lat} are top, bottom, and lateral melting rates, respectively. To simplify calculations, the model assumes all ice floes are circular, under this assumption the ice floe geometry and lateral melting of ice floes are given by

Table 2

Classification of Ice Thickness Aligned With the World Meteorological Organization Sea Ice Grid Classifications

Category	Description
Ice free	0 mm
New ice	1 mm
Ice rind	<100 mm
Young ice	100–150 mm
Gray–white ice	150–300 mm
Thin first year ice	300–700 mm
Medium first year ice	>700 mm

Note. Ice thickness categories are defined using left-inclusive, right-exclusive intervals $[h_{\text{min}}, h_{\text{max}})$. For example, a thickness of exactly 150 mm belongs to the Gray–White Ice category defined by the interval [150, 300).

$$p = \pi L \quad (2)$$

$$s = \alpha L^2 \quad (3)$$

$$w_{\text{lat}} = 1.6 \cdot 10^{-6} \Delta T^{1.36} \quad (4)$$

$$\frac{ds}{dt} = w_{\text{lat}} p \quad (5)$$

where p is the ice floe perimeter, s is the ice floe surface area, and ΔT is the difference in ice and water temperatures.

2.3. Ice Floe Climatology

In order to characterize the variability in ice floe size across the Great Lakes, an analysis of the US National Ice Center (hereafter, NIC) ice data was carried out. As described above, the NIC data provides daily information in which floe size is binned into seven predetermined ice type categories (Table 1). Given the categorical nature of this data set, standard continuous statistics (e.g., mean) are inappropriate, as they may result in nonphysical values falling between defined classes. Therefore, the statistical mode, representing the most frequently observed floe size category, was selected as the most robust metric to characterize predominant ice conditions. For each year in the period 2010 to 2021, the frequency of each ice floe category at every grid point across the data set was determined. The statistical mode of ice floe size at each grid point was then calculated using the daily timeseries, resulting in an annual map of spatially-varying ice floe size. In instances of multi-modal distributions or ties within a grid cell, the tie was resolved by selecting the category with the higher lake-wide frequency for that specific year, ensuring the selection reflected the broader regional ice character. In addition, the annual maximum ice cover (AMIC) for each year was extracted from the NIC data. The AMIC is a common metric used to classify the intensity of the Great Lakes winter ice season. The combination of the annual ice floe size mode and AMIC are used to characterize the drivers of interannual variability in ice floe sizes. Finally, to create an overall ice floe size climatology (Figure 5), a multi-step stability analysis was performed. First, the dominant (mode) ice floe category was identified for each individual year at every grid cell. In this context, dominance is defined through both space and time: quantitatively, it refers to the statistical mode of the daily time series calculated independently at each grid pixel, rather than a lake-wide aggregation. Second, the multi-year (2010–2021) climatology was determined by selecting the category that appeared as the annual mode in the greatest number of years. In the event of a tie, where multiple categories were dominant in an equal number of years, the category with the higher cumulative frequency across those years was selected. This “mode of annual modes” approach yields a spatially varying climatology that represents the most persistent interannual ice condition, used for the modeling scenarios in Section 2.5.

2.4. Lateral Melting Analysis

The NIC data was also used to assess the potential impact of a variable ice floe size on lateral ice melting rates. While top and bottom melting are driven by heat fluxes acting over the ice surface area (A) and are independent of floe geometry, lateral melting occurs along the floe perimeter. As shown in Equation 1, the lateral melting term is the only component scaled by the floe diameter (L). Consequently, lateral melting is the primary thermodynamic mechanism through which floe size variations influence ice volume evolution; specifically, smaller floes result in a higher perimeter-to-area ratio, thereby amplifying the contribution of lateral melting relative to top and bottom melting. To isolate this sensitivity, we applied Equation 1 analytically rather than utilizing full model output. By calculating the lateral melting coefficient ($\frac{\pi H}{\alpha L}$), directly from observations, we evaluated the sensitivity of ice melt potential to floe size variability independent of other model dynamics. Two scenarios were carried out using data from the 2019 and 2020 winter seasons: (a) a constant floe size scenario ($L = 300$ m) representative of current operational forecasting for the Great Lakes, and (b) a variable ice floe size scenario based on daily NIC data. In each scenario, ice thickness (H) was derived from the NIC data as the lake-wide average across all ice-covered elements at each time step. For the second scenario, ice floe size was determined from the mode of the daily NIC data across all lakes and three tests were carried out using the median, lower-, and upper-bounds of the ice floe diameter range given in Table 1. Specifically, the sensitivity limits were defined as the category minimum (L_{min}),

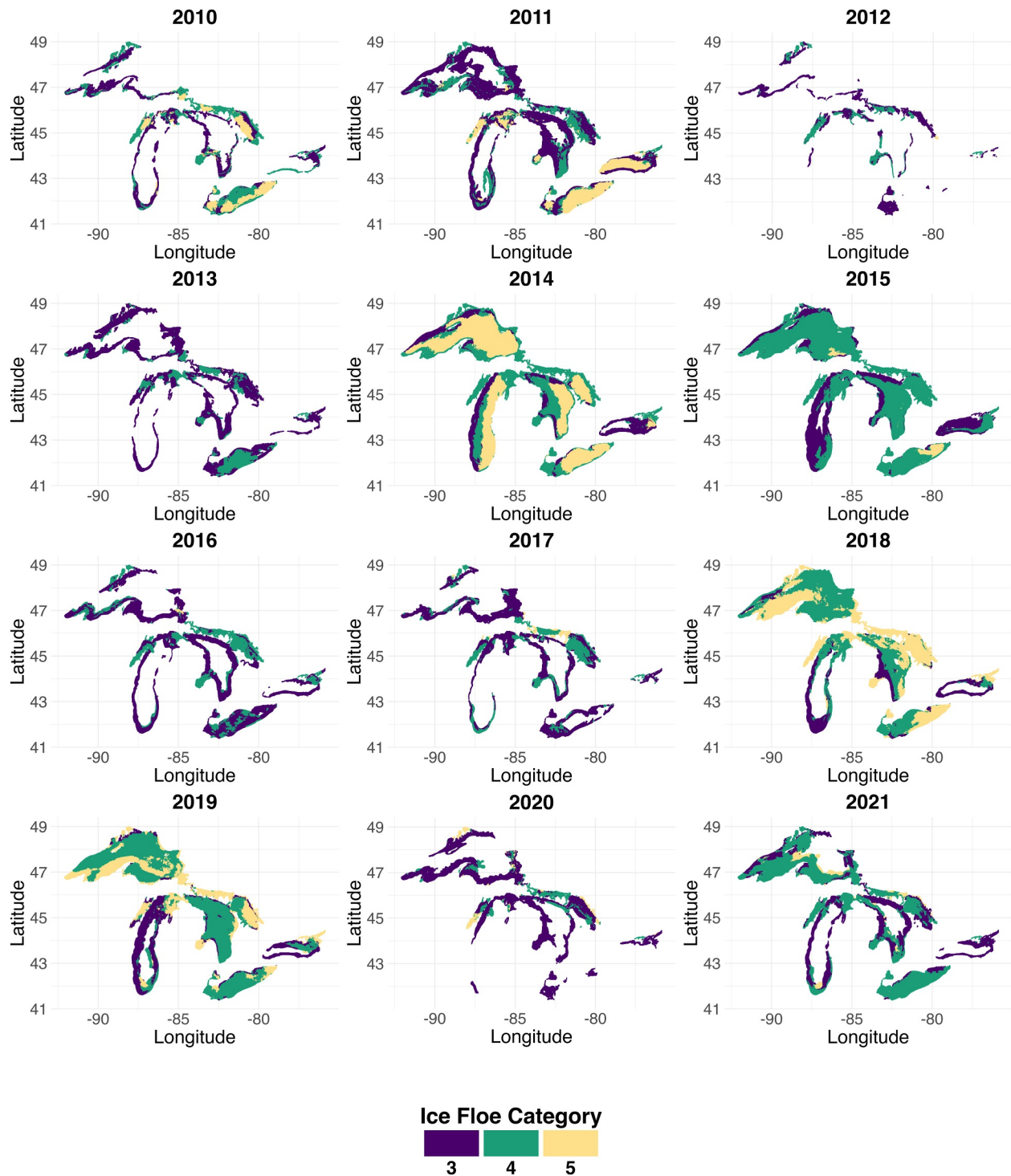


Figure 2. Annual ice floe diameter mode across the Great Lakes from 2010 to 2021. White indicates open water was the dominant mode for a location in that particular year.

median (L_{med}), and maximum (L_{max}) diameter for each floe size category. This procedure resulted in a time series of the lateral melting coefficient, facilitating an assessment of how floe size variability influences modeled ice melt dynamics. We then evaluated the relative impact of floe size variability by computing an amplification factor (ϕ), which represents the ratio of the lateral melting scaling parameter in the variable floe size cases to the constant 300 m reference case.

2.5. Model Experiments

Hydrodynamic ice model simulations were conducted for two contrasting winter seasons. The first period, from 15 November 2018, to 15 May 2019, was characterized by a severe mid-season Polar Vortex event that drove extensive ice formation, reaching a lake-wide Annual Maximum Ice Cover (AMIC) of 80.9% (NOAA GLERL, 2020). The second period, from 15 November 2019, to 15 May 2020, represented a significantly milder season with regional air temperatures approximately 3.4°C above average and a historically low AMIC of only 19.5%. These periods are referred to hereafter as winters 2019 and 2020, respectively. The selection of the 2019 and 2020 winter seasons was primarily driven by the availability of the specific atmospheric forcing used in NOAA's operational systems. The High-Resolution Rapid Refresh (HRRR) model (v3), which drives the operational GLOFS, became available in the summer of 2018. Consequently, earlier years with strong climatic signals could not be simulated with consistent operational forcing.

For both periods, the lake models were initialized on November 15 of each year from archived GLOFS output and driven by the same atmospheric forcing used in operations—the NOAA High-Resolution Rapid Refresh (HRRR) model, a 3-km data-assimilated implementation of the Weather Research and Forecasting (WRF) model (Benjamin et al., 2016; Dowell et al., 2022). For each winter, three scenarios were simulated: (a) a control case utilizing a constant ice floe size ($L = 300$ m), (b) a variable case implementing a spatially-varying ice floe size from the NIC described in Section 2.3, and (c) a daily variable case utilizing the time-varying daily NIC floe size analysis to assess the sensitivity of the model to temporal floe evolution. Hereafter, we refer to the three simulations as Control, Climatology, and Dynamic, respectively.

Model performance and the impact of ice floe parameterization was evaluated using satellite-derived products of ice concentration and thickness from the NIC and surface temperature from the GLSEA. Model skill was quantified using the Root Mean Square Error (RMSE) when continuous observational data were available (i.e., ice concentration and surface temperature). For ice thickness, where observational data are limited to broad categorical bins, we calculated the Root Mean Square Difference (RMSD) between the Control and Climatology simulations to quantify the sensitivity of the model physics to the floe size parameterization, rather than strictly defining model error. Finally, because NIC ice thickness is provided in ordered categories of varying range, a categorical accuracy assessment was conducted to evaluate model performance. Modeled ice thickness values were first interpolated to the NIC grid and then binned into the same categorical classes as defined by the NIC product. For each time step and grid location, the model and NIC category values were compared and summarized in a confusion matrix for each lake. From this, the total (exact match) accuracy was computed to represent the proportion of perfectly predicted values. Additionally, recognizing the ordinal structure of the categories, we calculated the proportion of predictions that fell within ± 1 bin of the NIC classification, allowing for small category mismatches.

3. Results

3.1. Ice Floe Climatology

Analysis of the NIC gridded data sets from 2010 to 2021 reveals significant spatial and temporal variability in ice floe diameters (L) across the Great Lakes. Spatial maps of ice floe size mode indicate notable differences in predominant floe sizes each year (Figure 2). Within each lake, the floe diameter mode fell within categories 3 to 5, corresponding to sizes ranging from 20 to 2,000 m (Table 1). In most years, coastal ice is dominated by category 3 (20–100 m) floes. Offshore ice, when present, generally ranges between 100 and 2,000 m (categories 4 and 5). A strong inverse correlation ($r = -0.868$, $p < 0.01$) was found between the normalized frequency of Category 3 and Category 5 floes (Figure 3). This statistical relationship reflects the physical trade-off between thermodynamic consolidation and dynamic fracturing processes. Severe winters promote the thermodynamic growth and consolidation of ice into vast, cohesive sheets (Category 5). In contrast, milder winters result in thinner ice cover that is more susceptible to wave-induced fracture, thereby limiting development to smaller, fragmented floes (Category 3). This pattern is clearly observed in the mild years of 2012, 2013, 2016, 2017, and 2020 (Figure 3). Notably, the GLOFS operational default value of 300 m, equivalent to category 4, is only dominant in 4 out of the 12 years investigated (Figures 2 and 4).

The relationship between winter intensity and ice floe size is revealed when the annual histogram of ice floe categories is compared to the AMIC, which illustrates the conditional dependence of ice floe size on the severity

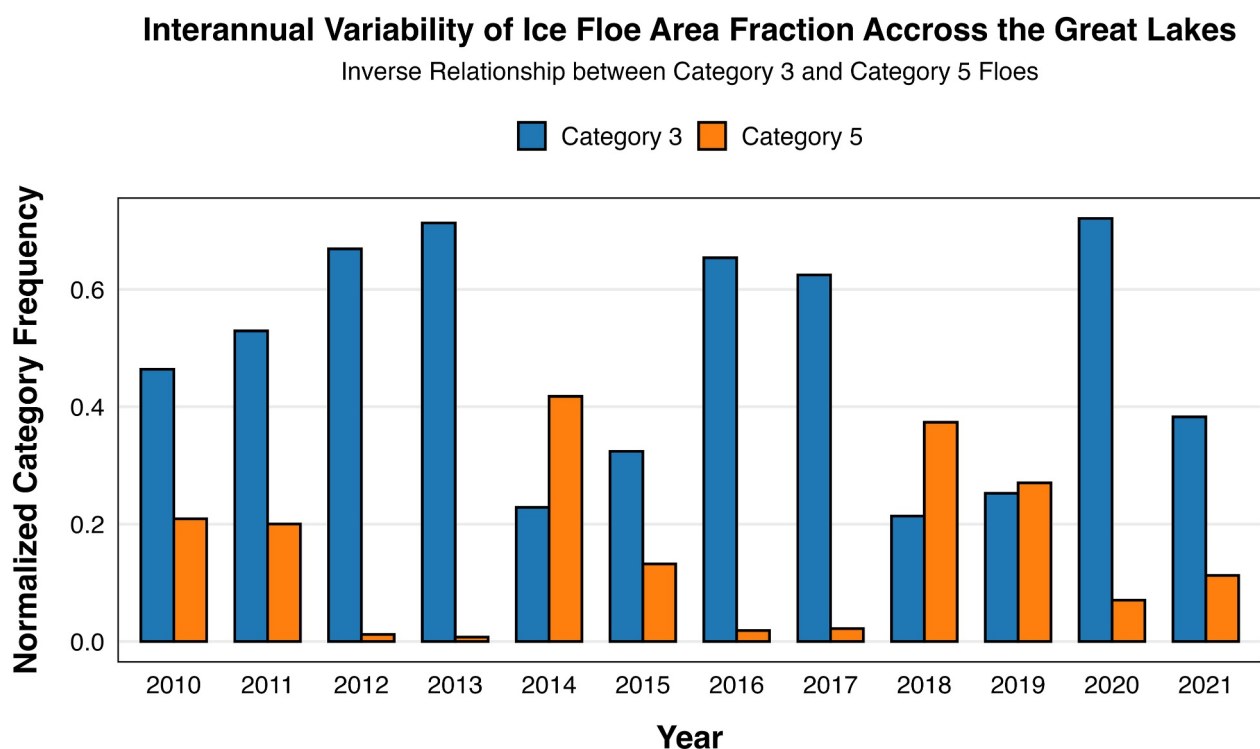


Figure 3. Temporal variation in the abundance of ice floe categories 3 (blue) and 5 (orange) across the Great Lakes.

of winter conditions across the Great Lakes (Figure 4). Historical data indicate that smaller ice floes (categories 3 and 4) are more prevalent during milder winters when ice cover is below 50%. In contrast, larger ice floes (category 5) dominate in most colder winters, where ice cover exceeds the 50% threshold. Category 4 ice floes are proportionally consistent across years with two notable exceptions: 2015, an extremely cold winter where average air temperatures (December–February) across the Great Lakes region were approximately 7.0°F (3.9°C) below the 1991–2020 average (NOAA NCEI, 2024), leading to significant ice cover, and 2020, an anomalously warm year with regional winter air temperatures around 6.1°F (3.4°C) above average, resulting in very little ice cover. When the mode of ice floe category is analyzed across all years, category 4 is the most prevalent for all lakes except Lake Ontario, which generally experiences very little ice cover in most years (Figure 5).

3.2. Impact on Lateral Melting Potential

The relationship between ice evolution and floe size is examined by calculating the lateral melting rate potential using Equations 1 through 5 across all the Great Lakes for two scenarios. Scenario (i) represents the operational forecasting standard, which employs a constant floe diameter of 300 m. In contrast, scenario (ii) incorporates three cases with variable floe sizes derived from the NIC climatology: the Median case, and the Lower Bound and Upper Bound cases (representing the minimum and maximum floe sizes within each NIC category, respectively). A significant discrepancy emerges when comparing the lateral melting coefficients obtained from these two scenarios (Figure 6). In both the high and low ice years examined, 2019 and 2020, respectively, the constant floe size scenario resulted in a steady increase in the lateral melting coefficient throughout the season. This rise was driven primarily by the prescribed increase in ice thickness (H) from the NIC data set. Conversely, the three cases utilizing variable floe sizes exhibited fluctuations in the lateral melting rate across the season generally exceeding the melting rates observed under the constant 300 m scenario. Specifically, the variable floe size scenarios exhibit significantly higher melt potentials in the early months of the ice season. During the Early Season (December–February) of 2019, the peak lateral melting coefficient was amplified by a factor (ϕ) ranging from 3 to 15 (Upper to Lower Bound) relative to the constant case. This amplification peaked earliest in the Upper Bound case (3.0x on 4 December 2018), followed by the Median case (5.0x on 21 December 2018), and finally the Lower Bound case, which reached a maximum of 15.0x on 8 January 2019. This elevated melting persists until February

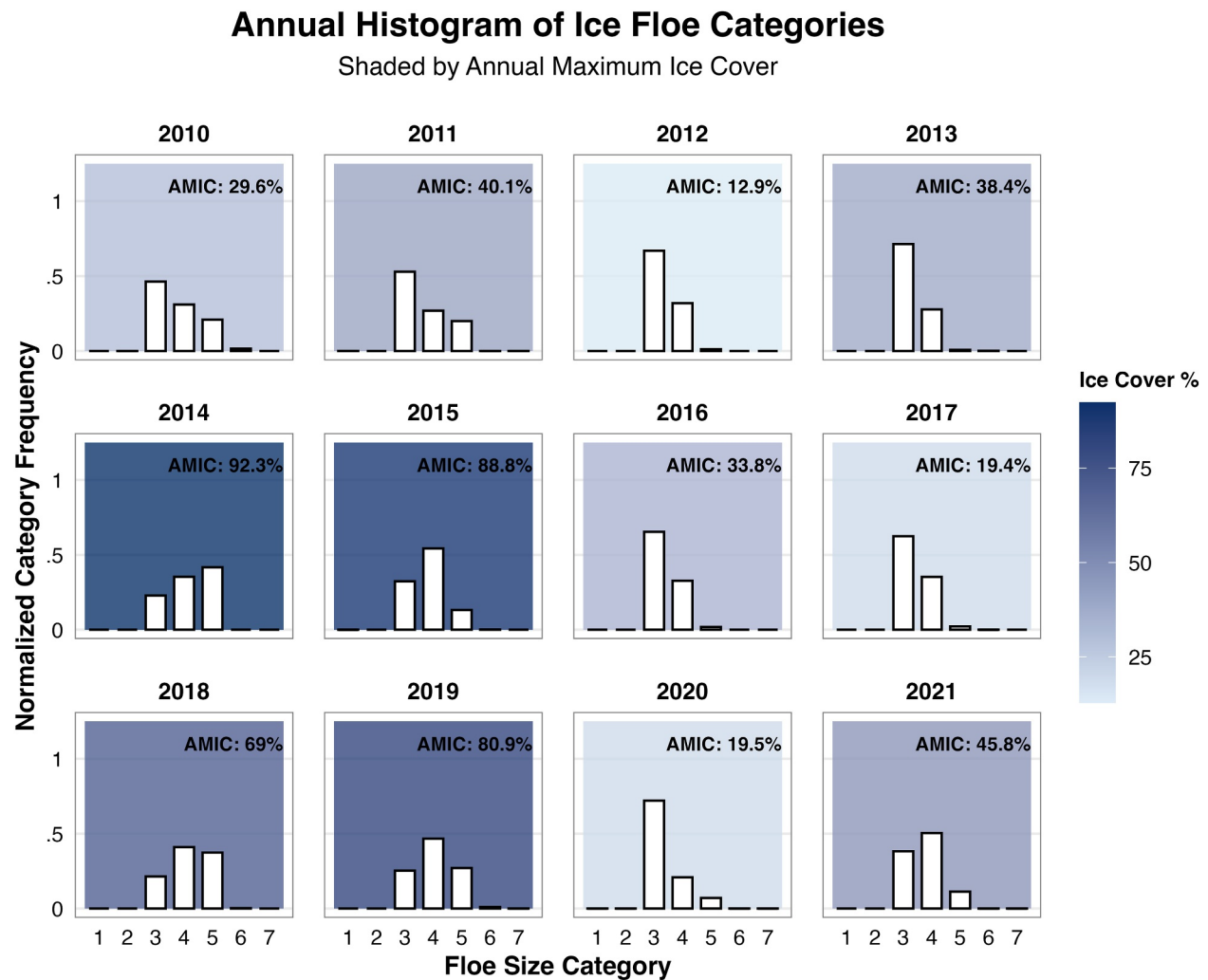


Figure 4. Annual histogram of lake-wide ice area fraction for ice floe categories from 2010 to 2021. Shading for each year is scaled by the Annual Maximum Ice Cover (AMIC) as an indicator of winter severity.

or March, after which the variable floe-size melt rates generally decline below the constant floe-size rate. An exception to this trend was observed in winter 2020, where the Lower Bound scenario maintained elevated melt rates throughout the season due to the persistence of smaller ice floes. While ice concentrations decline toward the end of the ice season, NIC-reported thickness continues to increase—reflected in the rising lateral melt rates under the constant floe size scenario. Meanwhile, the smaller floe sizes observed in April and May contribute to a secondary rise in lateral melt rates for the variable floe size cases. Notably in May 2020, the Lower-Bound amplification factor rebounded to 5.8, reflecting the rapid decay of ice into smaller floes during the spring melt. When considering the impact of a variable lateral melt rate on ice growth, the temporal evolution observed suggests that simulations incorporating variable floe sizes would produce thinner ice and potentially result in an earlier onset of spring melt compared to simulations with a constant floe size. Overall, the range of lateral melt rates across the variable floe-size scenarios ($L_{\max}$ to $L_{\min}$) is comparable to the full range observed in the constant floe size case. However, the constant 300 m floe size fails to capture the seasonal variability in the lateral melt coefficient, particularly late in the 2020 ice season, when the Lower-Bound case exhibits melt rates that far exceed those associated with a fixed floe size of 300 m.

3.3. Model Sensitivity to Ice Floe Parameterization

Model simulations with a variable floe size parameterization (Climatology and Dynamic) resulted in generally lower ice concentrations for each lake than the Control case with a constant floe size, with the largest differences

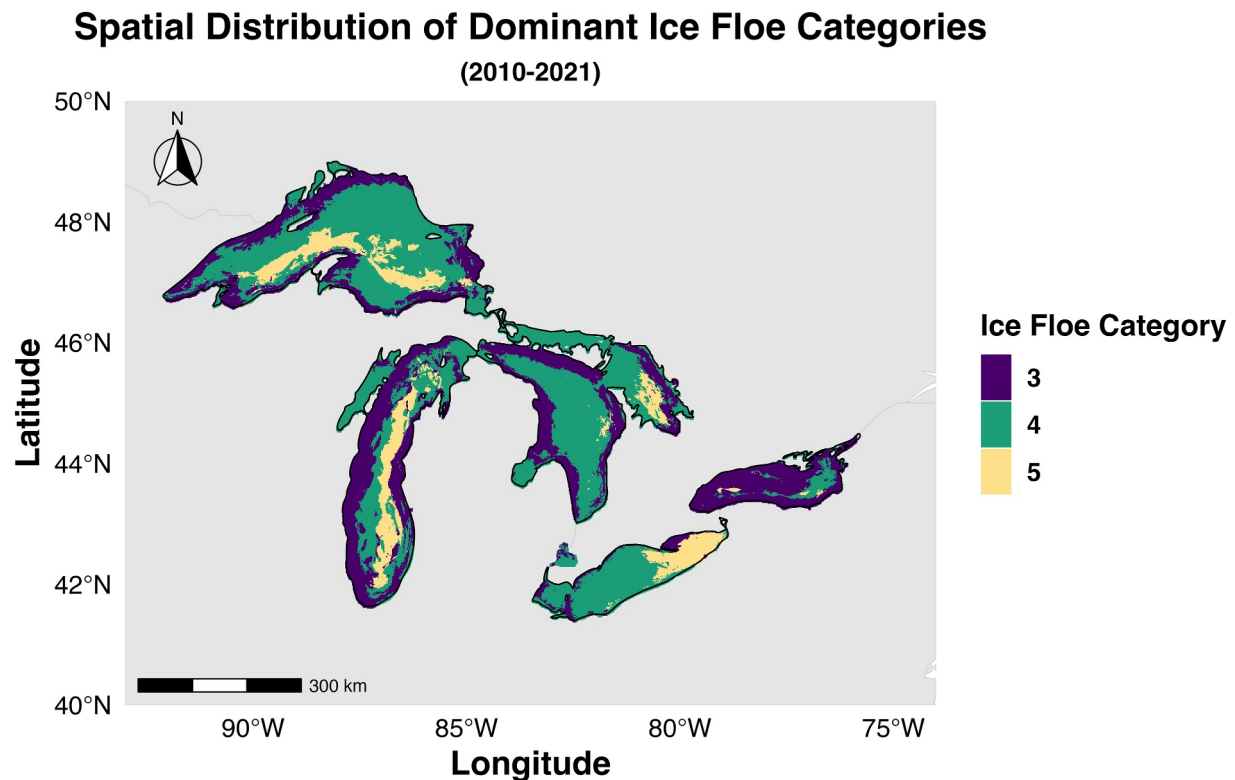


Figure 5. Ice floe size climatology (2010–2021) showing the *mode of annual modes*. Each grid cell represents the floe category most frequently dominant across all years. Ties were resolved by selecting the category with the higher cumulative frequency, ensuring representation of the most persistent floe type over the record (2010–2021).

occurring at the end of the ice season (Figure 7). This behavior aligns directly with the lateral melting analysis, which demonstrated that variable floe size parameterizations generate significantly higher melt rates during the first half of the winter season. These elevated early-season melt rates effectively limit ice accumulation, resulting in a reduced total ice volume that is primed for a more rapid decline in the spring, thereby accelerating the late-season melt-off compared to the Control simulations. In many cases, but not all, this resulted in better model performance when compared to the NIC concentrations. For Lake Erie, the Climatology simulation exhibited a mean bias of 5.45% in 2019 and 6.36% in 2020, while the Dynamic simulation showed similar improvements with biases of 5.21% and 6.28% respectively. In contrast, the Control simulation showed a considerably higher bias of 9.55% and 8.20% in both years (Table 3).

In Lake Michigan–Huron, the Climatology mean bias was 2.96% in 2019 and decreased to 2.16% in 2020. The Dynamic simulation performed similarly, with biases of 2.99% and 2.18%, while the Control simulation showed higher biases of 5.74% and 4.43%, respectively. Lake Ontario presented the highest discrepancy in 2019, with a Climatology mean bias of 17.89%, a Dynamic bias of 17.85%, and Control bias of 21.06%. These values dropped significantly in 2020 to 5.64% (Climatology), 5.65% (Dynamic), and 8.15% (Control). Finally, in Lake Superior, the Climatology simulation had a mean bias of 16.07% in 2019 and 12.78% in 2020. The Dynamic case showed a slight improvement in 2019 with a bias of 15.45% and was comparable in 2020 at 12.73%. The Control simulation again showed the highest biases of 20.27% and 18.25% in the same years. Overall, Climatology simulations consistently produced lower mean biases than Control simulations, especially in a severe winter such as 2019 when ice concentrations are highest.

Following the trends in lateral melting potential and ice concentration, modeled ice thickness is decreased in all cases when a variable floe size parameterization (Climatology and Dynamic) is used compared to a constant floe size parameterization (Figure 8). RMSD quantifies the absolute magnitude of differences between the two model scenarios, highlighting the sensitivity of ice thickness simulations to variations in ice floe size representation. The results reveal that the Dynamic and Climatology simulations produced nearly identical changes to the mean ice

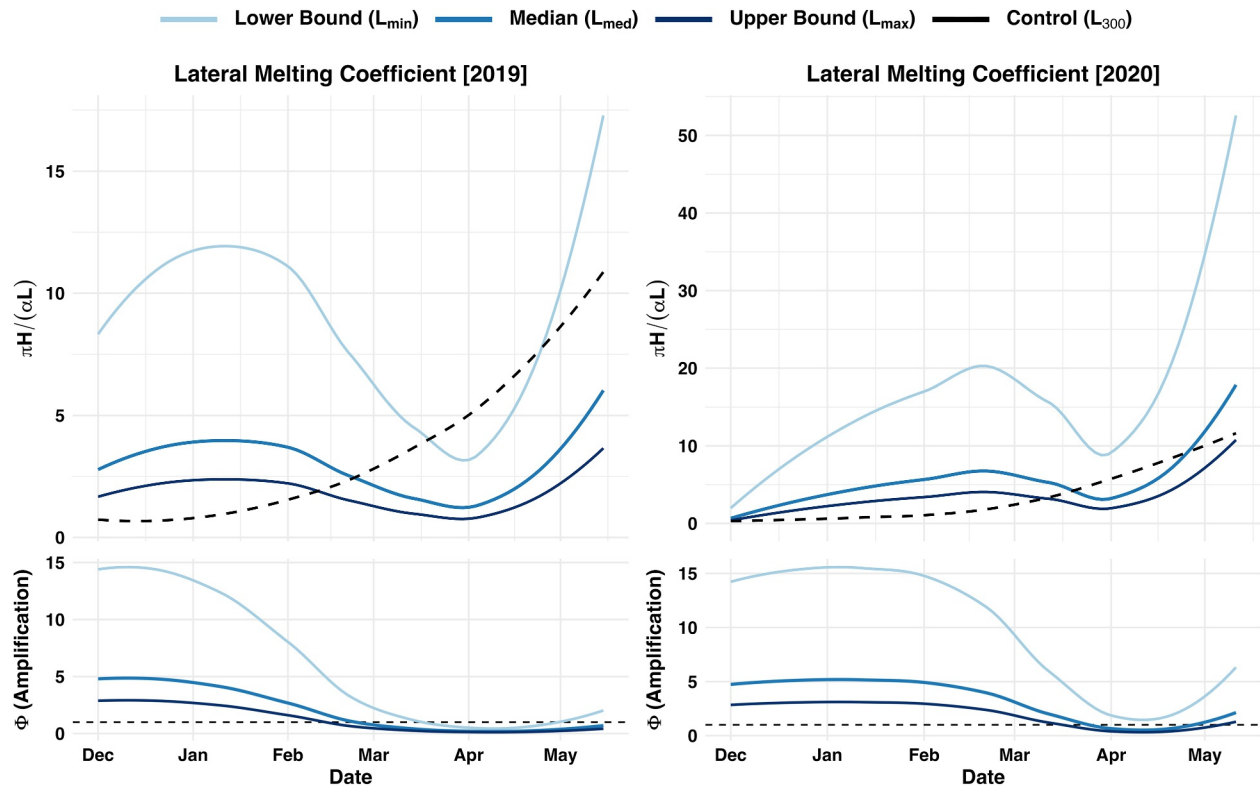


Figure 6. Comparison of lateral melt potentials between the constant and variable floe size scenarios for winter 2019 (left) and winter 2020 (right). The variable floe size scenario consists of three cases (lower, median, and upper bound modes).

state, with differences ranging from 1.8 to 7 cm in 2019 and <2.5 cm in 2020, with the largest differences found in Erie and Superior (Table 4). For instance, in the high-ice winter of 2019, the RMSD for Lake Erie was 0.055 m for both scenarios, and for Lake Michigan–Huron, it was 0.018 m for both. The largest deviation occurred in Lake Superior (0.063–0.069 m), yet the difference between the Dynamic and Climatology cases remained negligible (<0.006 m). While the NIC provides categorical ice thickness, a spatially-averaged thickness and upper and lower values of each NIC bin is plotted for reference against the modeled lake-average thicknesses (Figure 8). For most cases, the model is over-predicting and the decreased ice thickness in the variable floe size cases approaches the lake-averaged NIC. The exceptions are Superior in 2020 and both years for Michigan–Huron. However, treating the NIC thickness as a continuous value may bias interpretation, and thus a categorical analysis is more appropriate.

Rebinning the modeled thickness into the NIC categories for direct comparison, a multi-category ordinal assessment reveals that the model accuracy is markedly improved when a variable ice floe parameterization is used (Table 5). In terms of exact prediction, when the model prediction for a given space and time matches the NIC, the Climatology case improves over the Control for every lake and both years. The largest improvement occurs in Lake Superior in both years, moving from 41% to 67% in 2019 and 36% to 84% in 2020. Lake Ontario has the highest model accuracy of the lakes for both the Control and Climatology cases in both years. If model accuracy is instead assessed within 1 bin of the NIC ice category, overall the Control and Climatology predictive accuracy is higher compared to the exact match assessment. In most cases, the Climatology improves over the Control case, although the percent increase is smaller than found in the exact match accuracy. The Control case accuracy within 1 bin ranges from 78% to 99%, with Superior having the lowest accuracy and the highest accuracy found in both Lake Erie and Ontario. For the Climatology case, the accuracy within 1 bin ranges from 85% to 99%, where the lakes with the lowest and highest values are found again in Superior and Erie/Ontario, respectively. The only cases where the Control case performs better than the Climatology is for Michigan–Huron for the accuracy within 1 bin. However, the Climatology is only 4% and 2% lower than the Control case, with absolute accuracies at 87% and 95%, in 2019 and 2020, respectively.

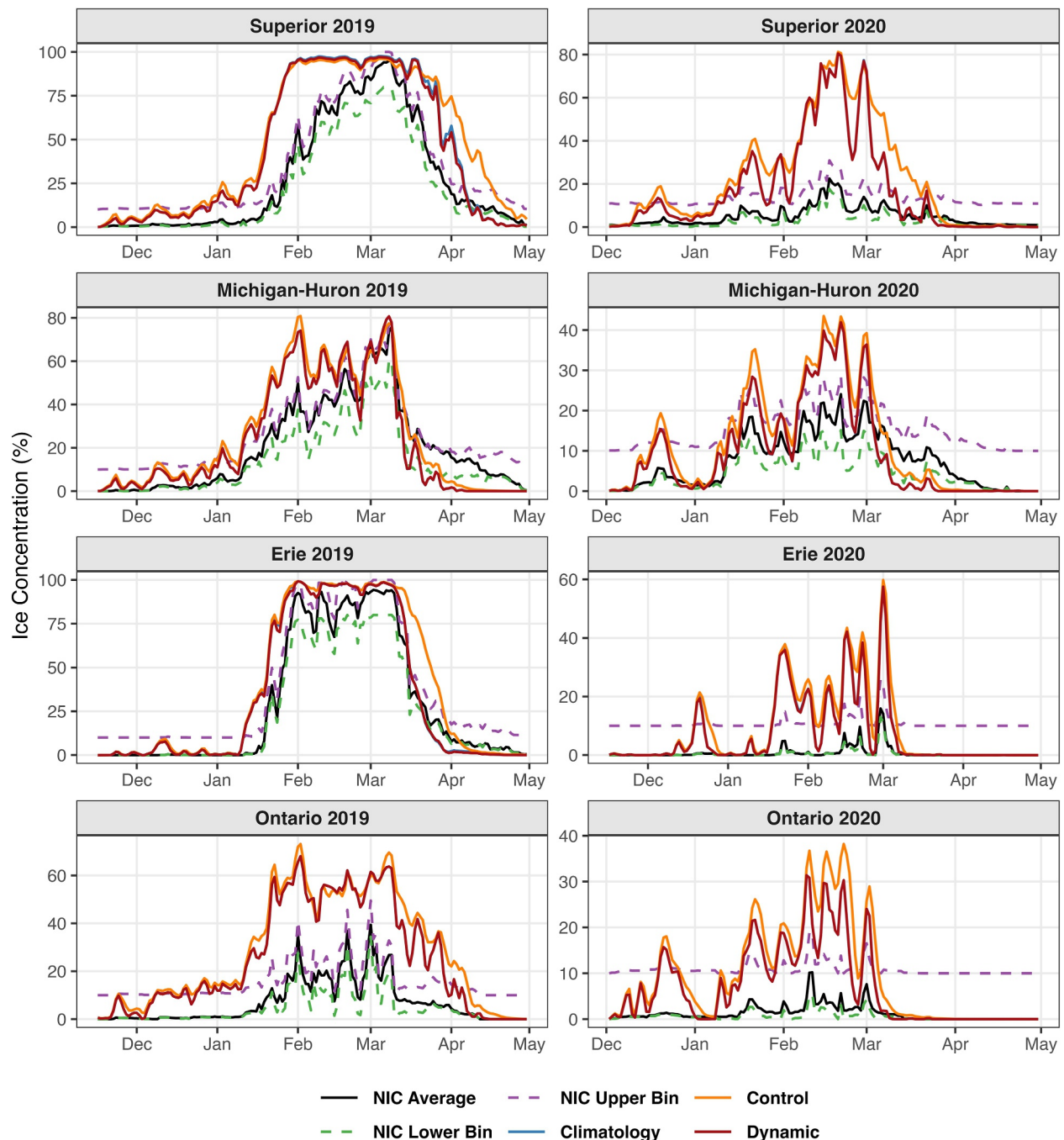


Figure 7. Lake-averaged ice concentrations [%] for 2019 and 2020 showing NIC Average (black, solid), NIC Lower Bin (green, dashed), NIC Upper Bin (purple, dashed), FVCOM Climatology simulation (blue), FVCOM Dynamic simulation (red) and FVCOM Control simulation (orange). Solid lines indicate mean values, while dashed lines represent NIC bin bounds.

Finally, lake-averaged surface temperature comparisons reveal that the model cases generally capture the seasonal trends observed in the GLSEA, though seasonal deviations are present; most notably a cold bias in winter and warm bias in spring for the deep lakes (Figure 9). While the Control simulations often exhibit marginally lower RMSE values (e.g., 1.11°C for Superior 2019 and 1.25°C for Michigan–Huron 2020), the variable floe size experiments remained highly competitive and, in some instances, improved accuracy (Table 6). Notably, the Dynamic simulation outperformed the Control in Lake Ontario during 2019 (1.57°C vs. 1.64°C) and Lake Erie during 2020 (1.33°C vs. 1.35°C). Overall, the differences in RMSE between the Climatology and Control runs are

Table 3
Mean Bias (%) and RMSE (%) of Modeled Ice Concentration Compared to NIC Data

Lake	Year	Bias (%)			RMSE (%)		
		Climatology	Dynamic	Control	Climatology	Dynamic	Control
Erie	2019	5.45	5.21	9.55	12.81	12.33	16.53
	2020	6.36	6.28	8.20	12.30	12.14	14.58
Michigan–Huron	2019	2.96	2.99	5.74	12.29	12.29	13.16
	2020	2.16	2.18	4.43	7.51	7.53	8.96
Ontario	2019	17.89	17.85	21.06	23.25	23.26	26.15
	2020	5.64	5.65	8.15	9.18	9.18	12.54
Superior	2019	16.07	15.45	20.27	23.75	23.14	26.63
	2020	12.78	12.73	18.25	22.09	22.07	27.02

minimal, generally within approximately 0.02–0.1°C, indicating that the ice floe climatology approach has a negligible impact on the accuracy of lake-averaged surface temperature simulations when compared to the Control, with both effectively capturing the seasonal evolution observed in GLSEA (Figure 9).

4. Discussion

Here, we demonstrate that ice floe size in the Great Lakes exhibits substantial spatial and temporal variability, counter to existing operational approaches that employ a constant value. The spatial distribution of ice floe sizes observed in the NIC record, particularly the prevalence of smaller floes in mild winters and larger floes in more severe seasons, highlight the importance of variable parameterization. These findings align with physical expectations that more severe winters produce thicker, more cohesive ice masses less prone to fracturing, while milder winters yield thinner, more fragmented floes. This suggests that ice floe parameterization may benefit from time-varying or seasonally-varying approaches, particularly given the observed trends in decreasing ice coverage and interannual variability found in the Great Lakes. Such parameterizations have already been made available as an experimental feature in the latest version of the CICE model (Hunke et al., 2024).

The lateral melt potential analysis shows that assuming a fixed 300 m floe size, as done in current operational systems, underestimates early-season melt rates and, in general, misses the seasonal variability in melt potential. By contrast, floe sizes derived from NIC data produce a melt coefficient that better reflects observed ice behavior, particularly those associated with small, pancake ice (or similar) at the start of the winter season and rapid melt observed at the end of winter. This mismatch between modeled and observed melt rates when using a fixed floe size may explain the commonly reported lag in simulated spring ice-out dates in GLOFS (Anderson et al., 2018; Fujisaki-Manome, Anderson, et al., 2020; Ye et al., 2019).

Across all lakes and both winters, simulations using the spatially varying floe size climatology consistently yielded lower mean bias in ice concentration compared to the Control case. The improvements were most pronounced during the high-ice year (2019), particularly in Lake Erie and Lake Superior. While this colder winter was characterized by larger floes similar to the Control setting (300 m), the extensive ice cover included vast marginal ice zones where smaller floes were prevalent. Because lateral melting scales inversely with floe diameter ($1/L$), the model is highly sensitive to these smaller floes. Consequently, the variable parameterization captured the rapid decay in these marginal zones, resulting in a larger absolute reduction in bias during the high ice year compared to the warm winter (2020), where total ice volume was minimal. These results emphasize the value of variable floe size parameterization for enhancing predictive skill across a range of ice conditions.

Similarly, modeled ice thickness was more consistent with NIC data when a variable ice floe size parameterization was used. Though absolute thickness comparisons are not possible due to the lack of direct ice thickness measurements and the categorical nature of the NIC data, an ordinal accuracy assessment shows that a variable ice floe size provides a better match with NIC categorical thickness. Most notably, Lake Superior showed significant improvement from 41% to 67% exact accuracy in 2019 and from 36% to 84% in 2020. These performance increases are notable especially when considering the relative ease of implementing climatological floe sizes into

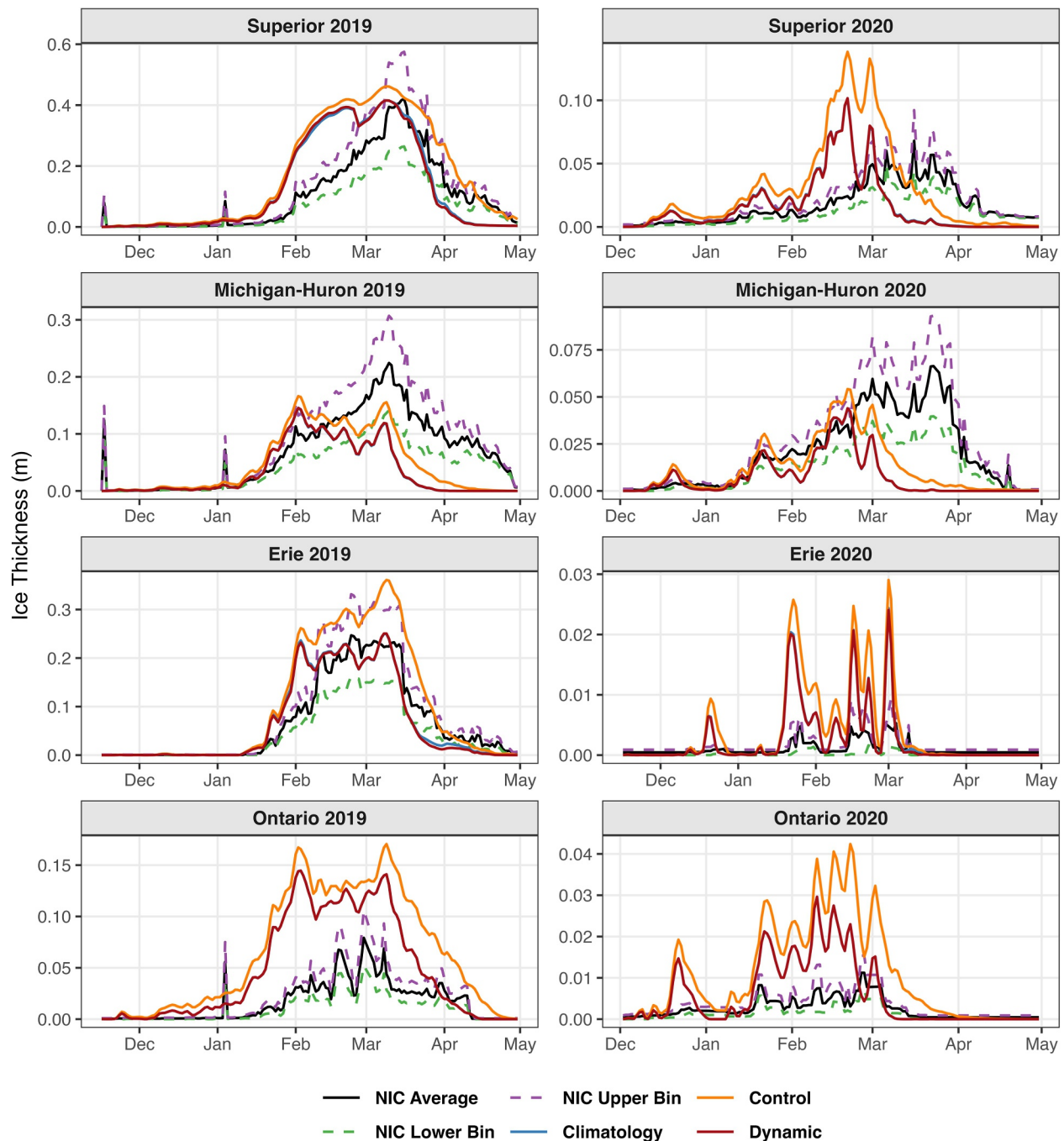


Figure 8. Lake-averaged ice thickness for 2019 and 2020 showing NIC Average (black, solid), NIC Lower Bin (green, dashed), NIC Upper Bin (purple, dashed), FVCOM Climatology simulation (blue), FVCOM Dynamic simulation (red), and FVCOM Control simulation (orange). Solid lines indicate mean values, while dashed lines represent NIC bin bounds.

operational models. In general, the results suggest that the constant floe size of 300 m used in operations contributes to overestimation of ice thickness early in the ice season, which leads to delayed melt in the spring.

Despite broad improvements in prediction of ice concentration and thickness, in some cases, the variable floe size led to early melt-off. For instance, in Michigan–Huron, while the variable ice floe size led to an increase in the number of exact matches with the NIC, the accuracy within ± 1 bin decreased slightly. This suggests that the floe size climatology resulted in a wider spread in model predictions and an increase in the extreme under- or over-

Table 4
RMSD Between Control and Climatology/Dynamic Scenarios for Ice Thickness

Lake	Year	RMSD (m) Climatology	RMSD (m) Dynamic
Erie	2019	0.055	0.055
	2020	0.003	0.0028
Michigan–Huron	2019	0.018	0.018
	2020	0.0073	0.0072
Ontario	2019	0.00229	0.00228
	2020	0.0072	0.0073
Superior	2019	0.063	0.069
	2020	0.021	0.022

predictions of thickness. To examine the impacts of temporal variability of floe size, we implemented a ‘Dynamic’ simulation that updated floe sizes daily using the NIC analysis. Surprisingly, the Dynamic case produced results nearly identical to the static Climatology case, with differences in ice thickness RMSD generally remaining below 1 cm (Table 4). This suggests that the primary driver of model error in the Control simulation is the spatial misrepresentation of floe size (i.e., applying a 300 m floe size in nearshore zones where small floes dominate), rather than the inability to capture high-frequency temporal fluctuations. While the daily assimilation of ‘real’ floe data (the Dynamic case) offers a marginal benefit in specific instances (e.g., Lake Ontario, 2019), the Climatology approach effectively corrects the systemic bias in the operational model without the need for a complex real-time data pipeline. Despite these improvements, discrepancies remain. A potential factor is that ice information from the NIC, particularly thickness, represents an amalgam of observations instead of an exact truth, which complicates model validation. Furthermore, ice floe size is just one compo-

ment of the modeled ice physics, and other processes should be considered in tandem. For instance, the current model configuration lacks comprehensive representation of nearshore ice dynamics, including land-fast ice and ice-wave interactions. Land-fast ice typically stabilizes nearshore regions, while wave-induced ice breakage enhances melt potential through increased exposure to water and air. Additionally, the inclusion of realistic snow cover, which significantly influences ice thermal properties and albedo, may also enhance model prediction. Meteorological uncertainties, particularly in surface air temperatures, which strongly influence ice cover variability, also remain critical factors contributing to model discrepancies.

Overall, our findings represent a significant advancement in ice prediction modeling for the Great Lakes, demonstrating the critical role of ice floe size parameterization in improving model performance. Integrating these advancements into operational forecasts is expected to provide tangible benefits for commercial navigation, hydropower generation, environmental management, and emergency response planning. Future research should prioritize the integration of remote sensing observations and explore model performance under data assimilative approaches.

Table 5
Model Accuracy for Exact Prediction of NIC Ice Thickness Category and Accuracy of Predicting the Category Within ± 1 Bin

			Superior	Michigan–Huron	Erie	Ontario
2019	Control	Accuracy	0.41	0.59	0.55	0.70
		Accuracy w/in 1 bin	0.78	0.91	0.92	0.97
	Climatology	Accuracy	0.67	0.79	0.64	0.91
		Accuracy w/in 1 bin	0.85	0.87	0.95	0.97
	Dynamic	Accuracy	0.65	0.77	0.69	0.90
		Accuracy w/in 1 bin	0.84	0.86	0.95	0.97
2020	Control	Accuracy	0.36	0.65	0.73	0.86
		Accuracy w/in 1 bin	0.86	0.97	0.99	0.99
	Climatology	Accuracy	0.84	0.91	0.77	0.98
		Accuracy w/in 1 bin	0.93	0.95	0.99	0.99
	Dynamic	Accuracy	0.82	0.91	0.79	0.98
		Accuracy w/in 1 bin	0.93	0.94	0.99	0.99

Note. Green shading indicates when the climatology or dynamic cases improve over the control case, purple shading is when the control case performs better.

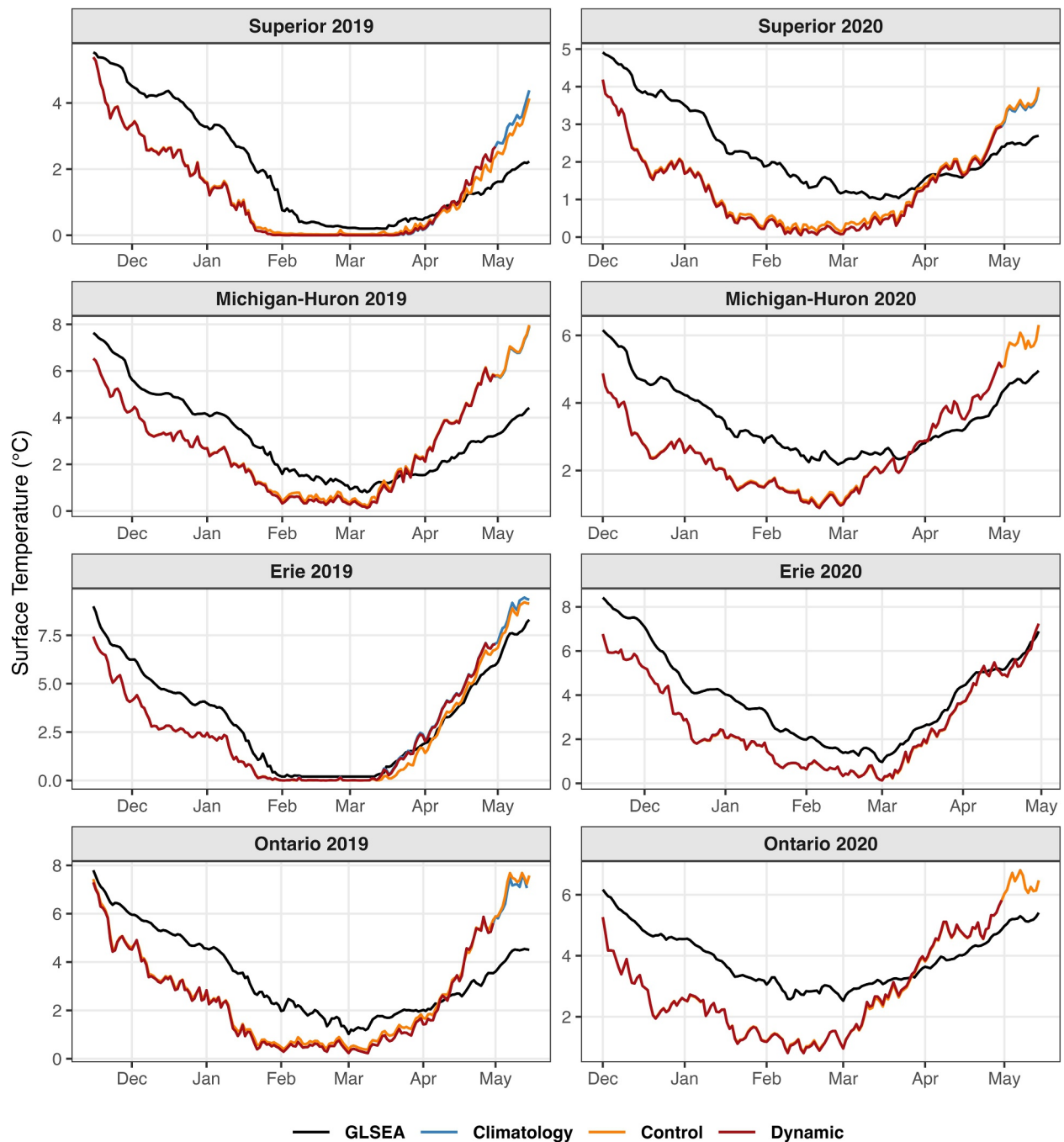


Figure 9. Lake-averaged surface temperature for 2019 and 2020 for the GLSEA (black), FVCOM Control simulation (orange), FVCOM Climatology simulation (blue), and FVCOM Dynamic simulation (red).

Table 6
RMSE (°C) of Modeled Surface Temperature Compared to GLSEA Observations

Lake	Year	Climatology RMSE (°C)	Dynamic RMSE (°C)	Control RMSE (°C)
Erie	2019	1.18	1.16	1.13
	2020	1.34	1.33	1.35
Michigan–Huron	2019	1.40	1.58	1.57
	2020	1.26	1.27	1.25
Ontario	2019	1.67	1.57	1.64
	2020	1.49	1.52	1.49
Superior	2019	1.17	1.14	1.11
	2020	1.26	1.29	1.22

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

All data used in this study are publicly available. Ice concentration, ice type, and ice floe size analyses are available from the U.S. National Ice Center (U.S. National Ice Center, 2023). Lake surface temperature data were obtained from the NOAA Great Lakes Surface Environmental Analysis (NOAA GLERL, 2023). Model simulations were conducted using the Finite-Volume Community Ocean Model (FVCOM v4.3; Chen et al., 2006) with the internally coupled CICE sea ice model (Hunke et al., 2017). Atmospheric forcing was provided by the High-Resolution Rapid Refresh model (Benjamin et al., 2016).

Acknowledgments

This is GLERL contribution number 2103, and CIGLR contribution number 1283. Kaitlin Pereira and Eric J. Anderson were partially funded by the National Oceanic and Atmospheric Administration and the Great Lakes Observing System (Grants: IOOS098/YR4-MODEL-03; IOOS030YR1-01).

References

- Anderson, E. J., Bechle, A. J., Wu, C. H., Schwab, D. J., Mann, G. E., & Lombardy, K. A. (2015). Reconstruction of a meteotsunami in Lake Erie on 27 May 2012: Roles of atmospheric conditions on hydrodynamic response in enclosed basins. *Journal of Geophysical Research: Oceans*, 120(12), 8020–8038. <https://doi.org/10.1002/2015jc010883>
- Anderson, E. J., Fujisaki-Manome, A., Kessler, J., Lang, G. A., Chu, P. Y., Kelley, J. G., et al. (2018). Ice forecasting in the next-generation great lakes operational forecast system (GLOFS). *Journal of Marine Science and Engineering*, 6(4), 123. <https://doi.org/10.3390/jmse6040123>
- Anderson, E. J., & Schwab, D. J. (2017). Meteorological influence on summertime baroclinic exchange in the Straits of Mackinac. *Journal of Geophysical Research: Oceans*, 122(3), 2171–2182. <https://doi.org/10.1002/2016jc012255>
- Anderson, E. J., Schwab, D. J., & Lang, G. A. (2010). Real-time hydraulic and hydrodynamic model of the St. Clair River, Lake St. Clair, Detroit river system. *Journal of Hydraulic Engineering*, 136(8), 507–518. [https://doi.org/10.1061/\(asce\)hy.1943-7900.0000203](https://doi.org/10.1061/(asce)hy.1943-7900.0000203)
- Anderson, E. J., Stow, C. A., Gronewold, A. D., Mason, L. A., McCormick, M. J., Qian, S. S., et al. (2021). Seasonal overturn and stratification changes drive deep-water warming in one of Earth's largest lakes. *Nature Communications*, 12(1), 1688. <https://doi.org/10.1038/s41467-021-21971-1>
- Anderson, E. J., Tillotson, B., & Stow, C. A. (2024). Indications of a changing winter through the lens of lake mixing in Earth's largest freshwater system. *Environmental Research Letters*, 19(12), 124060. <https://doi.org/10.1088/1748-9326/ad8ee0>
- Bai, P., Wang, J., Chu, P., Hawley, N., Fujisaki-Manome, A., Kessler, J., et al. (2020). Modeling the ice-attenuated waves in the Great Lakes. *Ocean Dynamics*, 70(7), 991–1003. <https://doi.org/10.1007/s10236-020-01379-z>
- Bai, X., Wang, J., Schwab, D. J., Yang, Y., Luo, L., Leshkevich, G. A., & Liu, S. (2013). Modeling 1993–2008 climatology of seasonal general circulation and thermal structure in the Great Lakes using FVCOM. *Ocean Modelling*, 65, 40–63. <https://doi.org/10.1016/j.ocemod.2013.02.003>
- Bateson, A. W., Feltham, D. L., Schröder, D., Hosekova, L., Ridley, J. K., & Aksenov, Y. (2020). Impact of sea ice floe size distribution on seasonal fragmentation and melt of Arctic sea ice. *The Cryosphere*, 14(2), 403–428. <https://doi.org/10.5194/tc-14-403-2020>
- Benjamin, S. G., Weygandt, S. S., Brown, J. M., Hu, M., Alexander, C. R., Smirnova, T. G., et al. (2016). A North American hourly assimilation and model forecast cycle: The rapid refresh. *Monthly Weather Review*, 144(4), 1669–1694. <https://doi.org/10.1175/MWR-D-15-0242.1>
- Cannon, D., Fujisaki-Manome, A., Wang, J., Kessler, J., & Chu, P. (2023). Modeling changes in ice dynamics and subsurface thermal structure in Lake Michigan–Huron between 1979 and 2021. *Ocean Dynamics*, 73(3), 201–218. <https://doi.org/10.1007/s10236-023-01544-0>
- Chen, C., Beardsley, R. C., & Cowles, G. (2006). An unstructured grid, finite-volume coastal ocean model (FVCOM) system. *Oceanography*, 19(1), 78–89. <https://doi.org/10.5670/oceanog.2006.92>
- Dowell, D. C., Alexander, C. R., James, E. P., Weygandt, S. S., Benjamin, S. G., Manikin, G. S., et al. (2022). The high-resolution rapid refresh (HRRR): An hourly updating convection-allowing forecast model. Part I: Motivation and system description. *Weather and Forecasting*, 37(8), 1371–1395. <https://doi.org/10.1175/WAF-D-21-0151.1>
- Durnford, D., Fortin, V., Smith, G. C., Archambault, B., Deacu, D., Dupont, F., et al. (2018). Toward an operational water cycle prediction system for the Great Lakes and St. Lawrence River. *Bulletin of the American Meteorological Society*, 99(3), 521–546. <https://doi.org/10.1175/bams-d-16-0155.1>
- Edson, J. B., Jampana, V., Weller, R. A., Bigorre, S. P., Plueddemann, A. J., Fairall, C. W., et al. (2013). On the exchange of momentum over the open ocean. *Journal of Physical Oceanography*, 43(8), 1589–1610. <https://doi.org/10.1175/JPO-D-12-0173.1>

- Fairall, C. W., Bradley, E. F., Hare, J. E., Grachev, A. A., & Edson, J. B. (2003). Bulk parameterization of air–sea fluxes: Updates and verification for the COARE algorithm. *Journal of Climate*, 16(4), 571–591. [https://doi.org/10.1175/1520-0442\(2003\)016<0571:BPOASF>2.0.CO;2](https://doi.org/10.1175/1520-0442(2003)016<0571:BPOASF>2.0.CO;2)
- Fairall, C. W., Bradley, E. F., Rogers, D. P., Edson, J. B., & Young, G. S. (1996). Bulk parameterization of air–sea fluxes for tropical ocean–global atmosphere coupled–ocean atmosphere response experiment. *Journal of Geophysical Research*, 101(C2), 3747–3764. <https://doi.org/10.1029/5JC03205>
- Fujisaki, A., Wang, J., Hu, H., Schwab, D. J., Hawley, N., & Rao, Y. R. (2012). A modeling study of ice–water processes for Lake Erie applying coupled ice–circulation models. *Journal of Great Lakes Research*, 38(4), 585–599. <https://doi.org/10.1016/j.jglr.2012.09.021>
- Fujisaki-Manome, A., Anderson, E. J., Kessler, J. A., Chu, P. Y., Wang, J., & Gronewold, A. D. (2020). Simulating impacts of precipitation on ice cover and surface water temperature across large lakes. *Journal of Geophysical Research: Oceans*, 125(5), e2019JC015950. <https://doi.org/10.1029/2019jc015950>
- Fujisaki-Manome, A., Mann, G. E., Anderson, E. J., Chu, P. Y., Fitzpatrick, L. E., Benjamin, S. G., et al. (2020). Improvements to lake-effect snow forecasts using a one-way air–lake model coupling approach. *Journal of Hydrometeorology*, 21(12), 2813–2828. <https://doi.org/10.1175/jhm-d-20-0079.1>
- Gao, G., Chen, C., Qi, J., & Beardsley, R. C. (2011). An unstructured-grid, finite-volume sea ice model: Development, validation, and application. *Journal of Geophysical Research*, 116(C8), C00D04. <https://doi.org/10.1029/2010jc006688>
- Hunke, E., Allard, R., Bailey, D. A., Blain, P., Craig, A., Dupont, F., et al. (2024). *CICE-consortium/CICE: CICE version 6.6.0 (6.6.0)*. Zenodo. <https://doi.org/10.5281/zenodo.14188742>
- Hunke, E., Lipscomb, W., Jones, P., Turner, A., Jeffery, N., & Elliott, S. (2017). *CICE, the Los Alamos sea ice model (No. CICE; 005315WKSTN00)*. Los Alamos National Lab (LANL). Retrieved from <https://www.osti.gov/biblio/1364126>
- Hunke, E. C., & Dukowicz, J. K. (1997). An elastic–viscous–plastic model for sea ice dynamics. *Journal of Physical Oceanography*, 27(9), 1849–1867. [https://doi.org/10.1175/1520-0485\(1997\)027<1849:aevpmf>2.0.co;2](https://doi.org/10.1175/1520-0485(1997)027<1849:aevpmf>2.0.co;2)
- Lin, Y., Fujisaki-Manome, A., & Anderson, E. J. (2022). Simulating landfast ice in Lake Superior. *Journal of Marine Science and Engineering*, 10(7), 932. <https://doi.org/10.3390/jmse10070932>
- Lin, Y., Fujisaki-Manome, A., & Wang, J. (2022). Recently amplified interannual variability of the Great Lakes ice cover in response to changing teleconnections. *Journal of Climate*, 35(19), 6283–6300. <https://doi.org/10.1175/jcli-d-21-0448.1>
- Mellor, G., & Blumberg, A. (2004). Wave breaking and ocean surface layer thermal response. *Journal of Physical Oceanography*, 34(3), 693–698. <https://doi.org/10.1175/2517.1>
- Mellor, G. L., & Yamada, T. (1982). Development of a turbulence closure model for geophysical fluid problems. *Reviews of Geophysics*, 20(4), 851–875. <https://doi.org/10.1029/RG020i004p00851>
- NOAA GLERL. (2023). Great lakes surface environmental analysis (GLSEA3) [Dataset]. *NOAA Great Lakes Environmental Research Laboratory*. Retrieved from <https://coastwatch.glerl.noaa.gov/glsea/glsea.html>
- NOAA Great Lakes Environmental Research Laboratory (GLERL). (2020). *Great Lakes ice cover database*. NOAA GLERL. Retrieved from <https://www.glerl.noaa.gov/data/ice/>
- NOAA National Centers for Environmental Information (NCEI). (2024). Climate at a glance: Regional time series. In *Great Lakes Region, Average Temperature, Dec-Feb for 2014-2015 and 2019-2020, with anomalies relative to 1991-2020 baseline*. Retrieved from <https://www.ncei.noaa.gov/access/monitoring/climate-at-a-glance/>
- Schwab, D. J., Leshkevich, G. A., & Muhr, G. C. (1999). Automated mapping of surface water temperature in the Great Lakes. *Journal of Great Lakes Research*, 25(3), 468–481. [https://doi.org/10.1016/s0380-1330\(99\)70755-0](https://doi.org/10.1016/s0380-1330(99)70755-0)
- Shiklomanov, I. A. (1993). World freshwater resources. In P. H. Gleick (Ed.), *Water in crisis: A guide to the world's fresh water resources*. Oxford University Press.
- Smagorinsky, J. (1963). General circulation experiments with the primitive equations: I. The basic experiment. *Monthly Weather Review*, 91(3), 99–164. [https://doi.org/10.1175/1520-0493\(1963\)091<0099:GCEWTP>2.3.CO;2](https://doi.org/10.1175/1520-0493(1963)091<0099:GCEWTP>2.3.CO;2)
- Thorndike, A. S., Rothrock, D. A., Maykut, G. A., & Colony, R. (1975). The thickness distribution of sea ice. *Journal of Geophysical Research*, 80(33), 4501–4513. <https://doi.org/10.1029/JC080i033p04501>
- U.S. National Ice Center. (2023). Great Lakes daily ice analysis [Dataset]. *U.S. National Ice Center*. Retrieved from <https://usicecenter.gov/Products/GreatLakesHome>
- Wake, A., & Rumer Jr, R. R. (1983). Great Lakes ice dynamics simulation. *Journal of Waterway, Port, Coastal, and Ocean Engineering*, 109(1), 86–102. [https://doi.org/10.1061/\(asce\)0733-950x\(1983\)109:1\(86\)](https://doi.org/10.1061/(asce)0733-950x(1983)109:1(86))
- Wang, J., Bai, X., Hu, H., Clites, A., Colton, M., & Lofgren, B. (2012). Temporal and spatial variability of Great Lakes ice cover, 1973–2010. *Journal of Climate*, 25(4), 1318–1329. <https://doi.org/10.1175/2011jcli4066.1>
- Wang, J., Hu, H., Schwab, D., Leshkevich, G., Beletsky, D., Hawley, N., & Clites, A. (2010). Development of the Great Lakes ice–circulation model (GLIM): Application to Lake Erie in 2003–2004. *Journal of Great Lakes Research*, 36(3), 425–436. <https://doi.org/10.1016/j.jglr.2010.04.002>
- Wang, J., Kessler, J., Bai, X., Clites, A., Lofgren, B., Assuncao, A., et al. (2018). Decadal variability of Great Lakes ice cover in response to AMO and PDO, 1963–2017. *Journal of Climate*, 31(18), 7249–7268. <https://doi.org/10.1175/jcli-d-17-0283.1>
- Woolway, R. I., Anderson, E. J., & Albergel, C. (2021). Rapidly expanding lake heatwaves under climate change. *Environmental Research Letters*, 16(9), 094013. <https://doi.org/10.1088/1748-9326/ac1a3a>
- Wright, D. M., Posselt, D. J., & Steiner, A. L. (2013). Sensitivity of lake-effect snowfall to lake ice cover and temperature in the Great Lakes region. *Monthly Weather Review*, 141(2), 670–689. <https://doi.org/10.1175/mwr-d-12-00038.1>
- Xue, P., Pal, J. S., Ye, X., Lenters, J. D., Huang, C., & Chu, P. Y. (2017). Improving the simulation of large lakes in regional climate modeling: Two-way lake–atmosphere coupling with a 3D hydrodynamic model of the Great Lakes. *Journal of Climate*, 30(5), 1605–1627. <https://doi.org/10.1175/jcli-d-16-0225.1>
- Ye, X., Anderson, E. J., Chu, P. Y., Huang, C., & Xue, P. (2019). Impact of water mixing and ice formation on the warming of Lake Superior: A model-guided mechanism study. *Limnology and Oceanography*, 64(2), 558–574. <https://doi.org/10.1002/lno.11059>
- Yeo, A. J., Anderson, E. J., Jablonowski, C., Wright, D. M., Mann, G. E., Fujisaki-Manome, A., et al. (2024). Assessing the potential for medium-range ice forecasts in the Laurentian Great Lakes. *Water Resources Research*, 60(9), e2024WR037507. <https://doi.org/10.1029/2024wr037507>